Attention Symbiotic Neural Network for Hyperspectral Image Refined Classification Based on Relative Water Content Retrieval

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***Abstract*— Hyperspectral image (HSI) classification approaches achieve significant improvements with the proposal and application of deep learning algorithms. However, due to the end-to-end structure of the deep learning model, the exploration of intrinsic physical–chemical properties in HSI data is insufficient, which restricts the extraction of diagnostic features and impedes the improvement of intraclass classification performance. Moreover, the synergetic spectral–spatial feature extraction in deep learning model is limited owing to the difference between HSI spectral and spatial dimensions, which also hinders the refinement of performance. In order to mitigate these issues, an attention symbiotic neural network (ASNN) based on relative water content (RWC) retrieval (RWCR) is proposed for HSI refined classification in this article. ASNN is a multisupervised deep learning model that is able to extract spectral–spatial and biochemical features from the multilabel input data simultaneously. The augmented multilabel data, consisting of original spectral–spatial labels and RWC labels, are generated in the RWCR inversion model, which contains the proposed spectral index [red edge slope (RES)] calculation and the proposed adaptive grading algorithm. There are two critical parts in ASNN, soft band selection (SBS) module and dimensionality-varied feature extraction (DVFE) module, which are responsible for attention assignment and synergistic spectral–spatial feature extraction, respectively. The experimental results on real HSI data verify the effectiveness of RES, SBS, and DVFE in ablation studies. It is also demonstrated that ASNN has the capacity for improving intraclass and interclass accuracy in refined classification and providing a competitive advantage in comparison with several state-of-the-art methods.**

***Index Terms*— Attention symbiotic neural network (ASNN), hyperspectral image (HSI) classification, relative water content retrieval (RWCR).**

I. Introduction

HYPERSPECTRAL image (HSI) sensing is one of the most important technologies for Earth observation, which provides hundreds of contiguous bands with narrow bandwidths for the surface area of interest [1]. The constant development and application of remote sensing largely benefit the accessibility of HSI data for scientific research, such as environmental protection [2], geological mapping [3], precision agriculture [4], forestry, hydrology, and scene recognition. The exploitation of hyperspectral data is paramount in practical research [5], [6], where HSI classification is one of the most pervasive tasks.

The main aim of classification [7] is to assign each pixel of an HSI spatial area to a certain label, and the complexity of data structure is addressed primarily in classification. Hyperspectral data consist of high-resolution spectral and low-resolution spatial information, which provide spatial and much more detailed spectral signatures of objects. However, the huge amount of data causes complex dimensionality processing [8], lots of redundancy, and some noise in bands [9]. Moreover, the collection of HSI ground truth is difficult, expensive, and time-consuming, which has contributed to the lack of labeled samples. The small number of training samples leads to the decrease of classification accuracy as the dimensionality increases, which is the well-known Hughes phenomenon [10]. Therefore, extracting the most informative features from HSI data makes classification a challenging task. Over the past few years, numerous HSI classification algorithms have been proposed aiming at effective feature extraction and accurate classification results [11], [12]. After statistical model approaches, the machine learning approaches have made great progress, in which deep learning methods have outperformed most others.

Machine learning methods applied in classification achieved significant improvements and obtained accurate classifiers. Some methods such as principal component analysis (PCA) and independent component analysis (ICA) generally concerned about spectral bands [13], [14], focusing on dimensionality reduction and band selection, which were proven effective to alleviate data redundancy and preserve critical and informative bands. However, the detailed information was easily lost, and the structural features of data were not extracted for classification. In order to make full use of HSI information, the support vector machine (SVM) was widely used as an effective supervised machine learning model for HSI classification. The varieties of SVM models [15], [16] and other kernel-based approaches were proposed and achieved a better performance. As opposed to SVM, some unsupervised algorithms such as k-nearest neighbor [17] and Markov random field [18] were able to obtain a satisfactory performance without the labels [19]. Besides, other methods such as improved logistic regression [20], rotation forest (RF) [21], and improved

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Bayesian models [22] were noticeable in the literature. In order to make use of spatial information, the spatial–contextual classifiers were developed [23] and achieved better performance than spectral-only classifiers. In general, supervised methods based on spectral–spatial information tended to achieve more significant improvements than other classification methods. Nonetheless, the handcrafted features extracted in machine learning methods were not able to satisfy the practical application, and the inherently sophisticated organization of HSI data increased the difficulty of providing satisfactory results consistently. As machine learning approaches develop continuously, numerous deep learning approaches are proposed simultaneously.

Deep learning classification methods consist of multiple layers for feature extraction within the end-to-end framework. The hierarchical models, in which the higher layers extract more abstract features than lower layers, can extract the relationship between the source and the target automatically. Without human subjectivity and arbitrariness in feature extraction, the autolearned features have stronger representation and higher expression ability than handcrafted features. Chen *et al.* [24] proposed a deep learning model based on spectral–spatial information, namely, stacked autoencoder (SAE), to extract high-level and abstract features for HSI classification. The extraordinary performance of SAE demonstrated the availability of deep learning algorithms for hyperspectral issues. Subsequently, a stacked sparse autoencoder was proposed by Zhao *et al.* [25] for HSI classification, which was able to extract deep features from spectral or spatial feature space. With the development of deep learning architecture, the deep belief network (DBN) [26]–[28] was introduced into hyperspectral classification and presented a better performance. Chan *et al.* [29] proposed a simple deep learning baseline for classification and designed the PCANet to learn features extremely easily. Based on this, a nonlinear spectral–spatial network (NSSNet) was proposed by Pan *et al.* [30] and improved the classification performance further. However, these classification approaches could not effectively make use of spatial information due to the model structure limitation. To address the limitation, a convolutional neural network (CNN) was proposed [31], [32] and acquired a better performance than the other approaches [33]. CNN was able to incorporate HSI spatial and spectral information for naturally extracting informative features, which enhanced the ability of classification. Because of the superiority in feature extraction, many varieties of CNN were studied by researchers [34]–[36]. The varieties of CNN can be generally grouped into four categories according to the type of feature extraction. The first type of CNN is 1-D CNN [37], which extracts spectral features from bands and discards the spatial information. The 1-D CNN performs well and obtains a better performance than common statistical models in certain circumstances. In order to utilize the structural relationship of neighboring pixels and the global spatial information, the second type of CNN, namely, 2-D CNN was introduced into the hyperspectral community from computer vision (CV). Because of the difference of channel numbers between HSI and color image in CV, spectral dimensionality of HSI data requires reduction to fit the models [38], [39]. The drawback of 2-D CNN for hyperspectral classification tasks is that making HSI data similar to computer images loses a large number of spectral bands and corresponding properties. The extra features extracted from spatial space cannot cover the loss of spectral features. Trying to mitigate the loss of spectral and spatial information, the third-type dual-channel CNN [40], which consists of 1-D CNN and 2-D CNN modules, was proposed for classification. In dual-channel CNN, 1-D CNN and 2-D CNN are responsible for spectral and spatial feature extraction, respectively. Then, the spectral and spatial features are concatenated and taken as the input of a classifier. The dual-channel CNN based on spectral–spatial information performs better classification results than the above two types of CNN. Similarly, the last type 3-D CNN [41], which extracts features based on spectral–spatial information, is able to deal with global information and achieve significant classification performance. The 3-D CNN takes 3-D patches as input and extracts the spectral–spatial features simultaneously by 3-D kernels, which achieves significant performance and outperforms other CNNs [42]. Besides the abovementioned CNNs, some other algorithms such as GoogleNet, VGGNet [43], DenseNet [44], and ResNet [45] were introduced into HSI classification and obtained demonstrable effects in different tasks.

Although deep learning models outperform others for HSI classification, there are still two kinds of issues to be addressed.

First, current classification models achieve outstanding classification performance in overall accuracy (OA), whereas lacking the full exploitation of the inherent properties of HSI data impedes the improvement of intraclass classification performance.

Most of the deep learning models focus on increasing the overall interclass distance, which easily results in the loss of subtle critical features of small-sized classes. Some diagnostic signatures (such as biophysical–biochemical features) are easily covered by other powerful features. Consequently, the critical difference between similar objects is unavailable to improve intraclass performance in refined classification. Specifically, deep models learn the relationship between the sample end and the label end so that the parameters in models are mainly trained based on label information. As pure data-driven methods, the deep learning models may overemphasize the mathematical process and pay not much attention to the subtle physical–chemical features contained in spectra. That is, deep learning models primarily focus on statistical characteristics of spectral curves, such as geometry and morphology features, which reflect the reflection and radiation characteristics of ground objects from visible to infrared spectral domains [46], [47]. The statistical characteristics can ensure the maximum profit of network with the minimum loss and increase the average interclass distance as much as possible. However, due to the high spectral resolution, spectral curves contain not only the statistical characteristics but also the subtle diagnostic characteristics, such as biophysical or biochemical features, which are vanishing during iterative training of layers. The neglected subtle features that affected

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by the properties of sensors and targets are usually noteworthy and physically explicable. For instance, vegetation covers a large part of Earth's land areas proverbially, and HSI data, as the important observation data of Earth, usually contain various vegetation. Chlorophyll in vegetation leaf absorbs most of the spectral energy at the 450- and 650-nm wavelengths and reflects the majority of it at 540-nm wavelength. The spectral curve of the leaf has a reflection peak at 540 nm, which makes leaves to appear green. With the decrease of chlorophyll content and the increase of anthocyanin and carotenoid content, the reflection energy and the peak at 540 nm are decreased and flattened, respectively, making leaves present yellow [48]. Obviously, spectra can reveal the subtle changes of the concentration of trace elements in leaves due to the high spectral resolution of HSI. Conversely, we can retrieve biophysical and biochemical parameters from spectra by inversion models, which facilitates the precise analysis of target surface material and the extraction of subtle spectral difference in classification. The biophysical–biochemical features are able to improve the classification performance among intraclass ground objects, which are easily misclassified due to the deficiency of diagnostic features but distinguishable physically or chemically. Therefore, the use of prior knowledge and biophysical–biophysical parameters (such as chlorophyll fluorescence and water content [49]) of spectra is a supportive way to improve the performance of deep learning-based refined classification.

Second, feature extraction units in deep learning models can extract efficient spectral–spatial features for classification. Nevertheless, the synergy of spectral and spatial feature extraction processes in the multiple layer structure needs to be improved. Deep learning models applied in HSI classification are usually inspired by CV models and designed for spectral–spatial feature extraction [50], [51], which are practical for HSI data classification. These approaches have the capacity to extract the accurate features and outperform handcrafted methods to date [52], [53]. However, the synergy of spectral–spatial feature extraction is limited owing to HSI spectral–spatial dimensionality difference, which hinders the extensibility of deep model and availability of higher level semantic features for refined classification. Besides, some other deep models transferred from the CV community are not compatible well with HSI data because of the inefficient collaboration of spectral–spatial feature extraction. CV data (RGB and RGBA) usually contain three or four spectral channels and lots of spatial pixels, which are able to satisfy the demand of most visual tasks. On the contrary to high-resolution spatial and low-resolution spectral CV data, HSI data tend to increase the number of spectral channels for spectral analysis, which generates high spectral resolution data with low spatial resolution. Consequently, the deep learning models from CV are not able to process HSI data directly. These models generally stop the feature extraction early when the spatial dimensionality cannot be reduced anymore. The early stopping makes remained spectral information invalid for further feature extraction. Hence, a deep learning model with the ability of synergistic spectral–spatial feature extraction is able to improve the refined classification performance.

In order to tackle the aforementioned issues, we propose a new deep architecture called attention symbiotic neural network (ASNN) for HSI classification in this article. ASNN is a multisupervised classification model, which includes two symbiotic processes: 1) relative water content (RWC) classification and 2) general spectral–spatial classification. These two processes not only make contributions to the common module of RWC and spectral–spatial features extraction but also to the classification improvement of each other. In order to select the different informative bands for two processes, we build a soft band selection (SBS) module based on attention mechanism in ASNN, which generates different dataflow orientations with the global band information. The core part of ASNN is a dimensionality-varied feature extraction (DVFE) and fusion module, which is an improved work of the previous CNN-based model [54]. DVFE is specially designed for ASNN to process HSI data, and it can coordinate the spectral and spatial feature extractions and alleviate the computational cost by various feature maps. In order to fulfill the requirement of symbiotic processes in ASNN, we also label all ground objects with RWC information to augment the original HSI data into multilabel data in the proposed inversion model. The feasibility and effectiveness of the proposed approaches are demonstrated in experiments.

In this regard, the main innovative contributions of ASNN proposed in this article can be summarized as follows.

1) A designed symbiotic architecture constrained by physical–chemical prior information is constructed. The designed end-to-end hierarchical architecture in ASNN has the ability to extract autolearned features from spectral–spatial and biochemical (RWC) information for refined classification. Moreover, in order to construct multilabel data sets consisting of original and RWC labels, a new RWC inversion model is introduced for ASNN. The inversion model is composed of two main RWC retrieval (RWCR) parts: the proposed spectral index [red edge slope (RES)] calculation and an adaptive grading algorithm for RWC labeling. The improvement of the classification performance of ASNN based on the RWC inversion model is demonstrated by experimental results.
2) A synergetic structure based on the DVFE module is introduced into ASNN for spectral–spatial feature extraction. The critical part in the synergetic structure is an attention mechanism-based SBS module, which is proposed for synergetic feature extraction and symbiotic attention assignment. SBS aims at selecting different informative bands by global attention generation and assigns them to different classification processes during synergetic feature extraction. So that the synergetic structure in ASNN is able to extract more representative spectral–spatial features and improve the symbiotic classification performance by respective attention masks. Moreover, the effectiveness of the SBS module in synergetic structure is demonstrated by experimental results.

The remainder of this article is organized as follows. Section II presents the proposed methodology and detailed algorithm implementation. Section III includes the description

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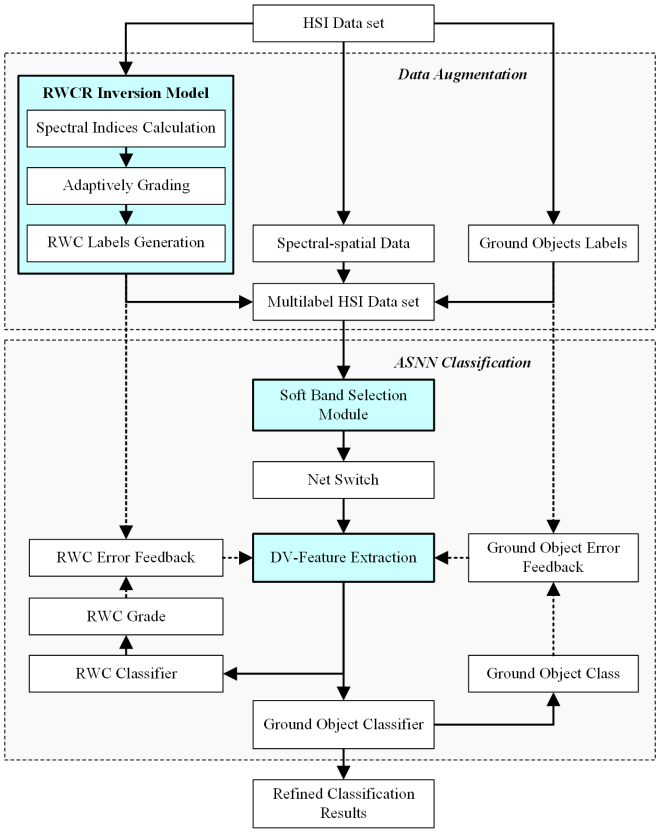


Fig. 1. Flowchart of implementing refined classification by ASNN.

of data sets, experimental settings, experimental results, and discussion. Finally, Section IV summarizes the conclusion.

II. Methodology of Proposed ASNN

As opposed to other typical CNN-based models, ASNN is a multisupervised classification model, requiring multiple label information of HSI data. The flowchart of refined classification implementation is shown in Fig. 1. There are two parts of the implementation procedure of ASNN classification. First, we conduct RWCR in the inversion model for data augmentation. We calculate the RWC values by spectral index and assign RWC labels to HSI data by adaptive grading algorithm. The operations in the inversion model combine RWC labels and original ground objects labels to generate the multilabel data set for ASNN. Second, ASNN takes the augmented data as input and extracts the features by the SBS module and multilayer DVFE module. After iterations, the refined classification results are output. According to the procedure of refined classification, this section is structured into two parts: RWCR for data augmentation in the inversion model and ASNN for HSI refined classification. The first part is a description of RWCR in the proposed inversion model, which generates the RWC labels by spectral index calculation and adaptive grading algorithm. The second part is a detailed description of ASNN, which contains the overview of ASNN and the elaboration of two key modules in ASNN: SBS and DVFE modules. We present the architecture of ASNN in detail

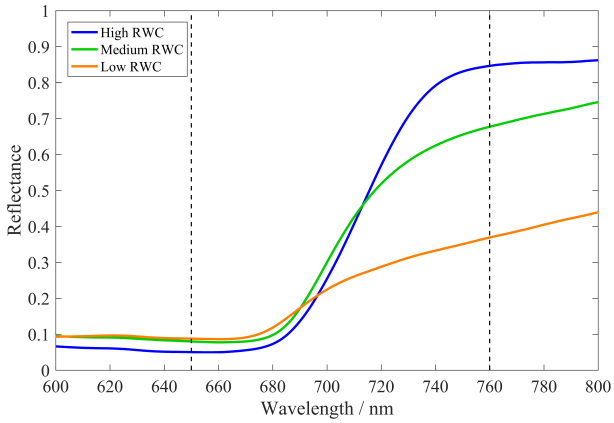


Fig. 2. Conceptual diagram of changes.

and elaborate on the structure and effectiveness of SBS and DVFE modules in theory.

A. Inversion Model of Relative Water Content Retrieval for Data Augmentation of ASNN

As a multisupervised model, ASNN requires multiple labels including original ground truth labels. That is, HSI data should be augmented by some other reinforcement information to satisfy the multilabel requirement of ASNN. In this article, biochemical information of RWC is selected for data augmentation.

RWC is a relative concept derived from the water content of vegetation. As we know, many land objects on Earth have their absorption and scattering properties that can be exhibited in the continuous narrow wavebands of hyperspectral curves. The spectral characteristics reflect their inherent biophysical or biochemical properties [55]. Due to the large coverage and sufficient quantities, the abundance of vegetation (such as various trees, grasses, and crops) in Earth observation is much more than that of any other ground objects. The spectral curve of vegetation has a typical morphological phenomenon between red and near-infrared (NIR) regions (650–760 nm wavelength), which is called the red-edge effect. It has special relevance to cell structure and water content in leaves [56]. We also researched on other objects (except vegetation) and observed an analogous phenomenon: the more water contained or the more competent to retain water, the more tremendous changes between red and NIR regions. The conceptual diagram is shown in Fig. 2. Consequently, we introduce RWC, the relative concept to vegetation water content, to describe this phenomenon.

The advantages of RWC information utilized by HSI classification are not only satisfying the data requirement of ASNN but also tackling some typical and intractable issues by biochemical information. For instance, some classification methods may misclassify objects such as plants and stones. However, it is easily distinguishable because of the water content difference, and the spectral features are able to reflect this difference [57]. The reason for misclassification is that the subtle spectral signature is ignored by hand-designed

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feature extraction models (machine learning approaches) or overwhelmed by conspicuous statistical features in data-driven models (deep learning approaches). ASNN based on RWC information is capable of increasing the conspicuousness of subtle spectral signature and enhancing the representation of diagnostic features. For multilabel data construction of ASNN, we build an inversion model to retrieve RWC and generate RWC labels by spectral index calculation and adaptive grading algorithm, respectively.

1) Spectral Index Calculation for Relative Water Content Retrieval: A set of spectral indices based on prior physical–chemical knowledge was proposed for RWCR in the literature. Spectral indices are not only effective and concise in RWC calculation but also with good portability and stability in RWC presentation. Researchers usually estimate biophysical and biochemical parameters based on spectral index approaches because they are conceptually simple, repetitive, and related to biophysical–biochemical characteristics closely. Besides, the calculation of spectral indices among bands can remove a certain influence of illumination.

Some spectral indices in the literature related to RWC are introduced, and the calculations of frequently used spectral indices are briefly described as follows.

Water Index (WI)

$$\mathrm{WI} = \frac{R_{900}}{R_{970}}. \tag{1}$$

Red Edge Normalized Difference Vegetation Index (NDVI_{750})

$$\mathrm{NDVI}_{750} = \frac{(R_{750} - R_{705})}{(R_{750} + R_{705})}. \tag{2}$$

Modified Red Edge Simple Ratio Index (MSR_{750})

$$\mathrm{MSR}_{750} = \frac{(R_{750} - R_{445})}{(R_{750} + R_{445})}. \tag{3}$$

Spectral Absorption Index (SAI_{650})

$$\mathrm{SAI}_{650} = \frac{(\lambda_{760} - \lambda_{650})R_{540} + (\lambda_{650} - \lambda_{540})R_{760}}{(\lambda_{760} - \lambda_{540})R_{650}} \tag{4}$$

where R and λ indicate the spectral reflectance values and the wavelength value, respectively, of each band. The subscript number is the wavelength.

In order to enhance the compatibility and stability of spectral indices and deep learning model, we proposed a new spectral index call RES to describe the water content changes between red and NIR regions. We select two key bands between red and NIR regions to simulate the red edge and estimate the RWC based on simulation. Instead of the slope at the inflection point of red edge [58], RES is capable of reducing the influence of subtle factors. The calculation of RES is given by

$$\mathrm{RES} = 2\sin\left(\frac{\Delta R_{760,650}}{\Delta \lambda_{760,650}}\right) \tag{5}$$

where R_{760} and R_{650} are the spectral reflectance values at 760-nm (λ_{760}) and 650-nm (λ_{650}) wavelength, respectively. $\Delta R_{760,650}$ and $\Delta \lambda_{760,650}$ are the difference of spectral reflectance and wavelength values between 760 and 650 nm, respectively.

The values of spectral reflectance are normalized into the range [0, 1]. For simplicity, (1) can be reformulated as

$$\mathrm{RES} = 2\sin(R_{760} - R_{650}). \tag{6}$$

All of these spectral indices are transferred from vegetation and applied to estimate RWC of ground objects. RES and other spectral indices are tested in comparison experiments, and the most sensitive and correlative spectral index is adopted for the adaptive grading algorithm to generate RWC labels.

2) Adaptive Grading Algorithm for RWC Labels' Generation: We can obtain the same amount of RWC values as ground truth labels by spectral index calculation. However, all of the RWC values are continuous in a range, which should be graded and labeled by an adaptive grading algorithm for data augmentation. The process of adaptive grading algorithm can be described as follows.

First, let O denote the output of spectral index calculation, and it can be presented as a cluster $O = \{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n\}$, where n is the number of ground objects, and $\boldsymbol{\alpha}$ refers to the RWC value vector of each object. Let m be the number of samples in each object, and $\boldsymbol{\alpha}$ can be described as $\boldsymbol{\alpha} = \{\upsilon_1, \upsilon_2, \ldots, \upsilon_m\}$ with υ being the value of RWC. According to the task requirement of kRWC labels, we need to divide n ground objects into $k(k \le n)$ clusters $\{C_i | i = 1, 2, \ldots, k\}$. Every cluster contains several kinds of objects and does not overlap each other.

Second, we define dist based on Hausdorff's distance dist_H [59] as the distance of two objects $\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j$, which is given by

$$\mathrm{dist}(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) = \mathrm{dist}_H(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) + \mathrm{dist}_C(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) \tag{7}$$

$$\mathrm{dist}_H(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) = \max(\mathrm{dist}_h(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j), \mathrm{dist}_h(\boldsymbol{\alpha}_j, \boldsymbol{\alpha}_i)) \tag{8}$$

$$\mathrm{dist}_C(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) = \frac{|\mathrm{avg}(\boldsymbol{\alpha}_i) - \mathrm{avg}(\boldsymbol{\alpha}_j)|}{\min(\mathrm{std}(\boldsymbol{\alpha}_i), \mathrm{std}(\boldsymbol{\alpha}_j))} \tag{9}$$

$$\mathrm{dist}_h(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_j) = \max_{\upsilon_i \in \boldsymbol{\alpha}_i} \min_{\upsilon_j \in \boldsymbol{\alpha}_j} \|\upsilon_i - \upsilon_j\|_2 \tag{10}$$

where dist_C is the calibration term of dist to avoid the impact of maximum or minimum values caused by sensor mode, atmospheric effects, or artificially labeling mistakes.

Third, we adopt a clustering algorithm AGNES to grade RWC values and label them on the basis of distance calculation. AGNES is a bottom-up hierarchical clustering algorithm, and the process of AGNES is shown in Algorithm 1.

After the generation of RWC labels, we can construct the multilabel data set based on RWC labels and original ground truth labels. ASNN takes the multilabel data sets as input and implements multiple classification tasks. Because of the abundant information in input data, ASNN is able to extract subtle, abstract, and representative signatures from spectral–spatial and biochemical RWC information by the symbiotic mechanism.

B. Attention Symbiotic Neural Network for HSI Refined Classification

Taking multilabel data as input, ASNN is a data-level fusion model for RWC and spectral–spatial information. From a

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Algorithm 1 AGNES Clustering

```
procedure AGNES(O, k)
  Initialize C
  for i = 1, 2, ..., n do
    α_i = O_i
    C_i = {α_i}
  end for
  get the total number q of clusters in
  C = {{α_1}, {α_2}, ... {α_n}}
  while q > k do
    get the clusters C = {C_1, C_2, ... C_q}
    calculate the distance matrix D of clusters
      for i = 1, 2, ..., q do
        for j = i + 1, ..., q do
          D(i, j) = dist(C_i, C_j)
        end for
      end for
    find the clusters C_i', C_j' with the minimum distance
    M(i', j')
    unite C_i' and C_j': C_i' = C_i' ∪ C_j'
    reset the C as {C_1, C_2, ... C_q-1}
    get the total number of clusters in C: q = q − 1
  end while
    for i = 1, 2, ..., q do
      for j = 1, 2, ..., t do
        get C_i = {α_1, α_2, ... α_t}
        relabel α_j with new grade label i
      end for
    end for
end procedure
```

standpoint of deep learning, the RWC information introduced as expert knowledge is able to enhance the decisive features, and the multisupervised model with extra labels contributes to accurate features and improvement of classification performance. From the other points of physics, RWC information is a kind of biochemical characteristic constraint, which increases the physically explicable of deep learning model to a certain extent.

1) Overview of Attention Symbiotic Neural Network: ASNN contains two main data processing channels: general spectral–spatial classification and RWC classification. The architecture of ASNN is shown in Fig. 3. There are two parts composed of the architecture. The upper part is the main architecture of ASNN, and the lower part is the specific illustration of DVFE, which is one of the major modules in ASNN.

In general, ASNN consists of five major components between the input and output layers. The input of ASNN is a raw HSI cube data that contain spectral–spatial information and consist of pixel-wise bands. Let $\mathbf{C} \in \mathbb{R}^{H\times W\times N}$ denote the HSI cube, where H, W, and N are the height, width, and number of spectral bands in HSI data, respectively. The first major component of ASNN is the patch data layer, which extracts patch data from HSI cube and feeds them into the next component. The patch data can be denoted by $\mathbf{X} \in \mathbb{R}^{M\times M\times N}$ similarly. The patch size $M \times M$ depends on the region selected from HSI cube, and the number of neighbor pixels in the region is a decisive factor for spatial information abundance. The second major component of ASNN is the SBS module based on the attention mechanism. In this module, two different soft selection criteria are applied for patch data according to different classification tasks. After SBS of data, the two corresponding feature maps are generated and taken as the input of net switch, the third major component. Each of the generated feature maps can be denoted as $\mathbf{B} \in \mathbb{R}^{M\times M\times N}$. The net switch module is responsible for selecting and feeding features into the fourth major component, the DVFE module. DVFE module is a critical part of extracting spectral–spatial and RWC features with computational cost reduction. DVFE module contains three different feature extraction stages along with three different dimensionality variations of feature maps. The first stage is the 3-D feature extraction, which includes multiple 3-D convolution layers and maxpooling layers. The 3-D operations are able to extract fused features from the input feature maps. The spatial and spectral dimensionalities of feature maps are reduced, and the 1-D vectors of features are generated. All the vectors are simplified features containing abundant spectral–spatial–RWC information with lower dimensionalities. The second stage is responsible for further feature fusion which assembles all the 1-D vectors into a 2-D feature map, and the fused 2-D feature map is sent to the third stage. The feature fusion module aims to collect global spectral–spatial information to an informative matrix and generate the input feature map for the next stage. The specific operation of feature fusion is that all 1-D vectors are combined in spatial axis keeping the band direction. The third stage is a 2-D feature extraction unit, where multiple layers constitute a general 2-D CNN. In this stage, numerous 2-D convolution and maxpooling operations compose the basic feature extraction module, and some other operations, such as activation layers and fully connected layers, are also applied. As a result of 2-D feature extraction, a vector is taken as the input of the classifier, which is the fifth major component of ASNN. The classifier is responsible for vector feature classification and outputs the results.

ASNN is a sequential iterative model (SIM) that two training processes are carried out successively and repeatedly. The training and testing processes of ASNN are similar to that of others. All of the parameters are updated in every iteration epoch. The model SIM can be formulated as

$$\mathrm{SIM} = t_1 P_1(\mathbf{X}) + t_2 P_2(\mathbf{X}) \tag{11}$$

where t_1 and t_2 are the running sequence of two processes $P_1(\cdot)$ and $P_2(\cdot)$, and $\mathbf{X}$ is the aforementioned input patch. Each of the processes includes forward feature extraction and backward parameters updating, and the forward process is given by

$$P(\mathbf{X}) = f_C(f_{\mathrm{DEFV}}(f_{\mathrm{AS}}(\mathbf{X}))) \tag{12}$$

where $f_{\mathrm{AS}}(\cdot)$, $f_{\mathrm{DEFV}}(\cdot)$, and $f_C(\cdot)$ represent the attention-based SBS, DEFV, and the classifier, respectively. The updating of $P(\mathbf{X})$ after backward propagation can be formulated

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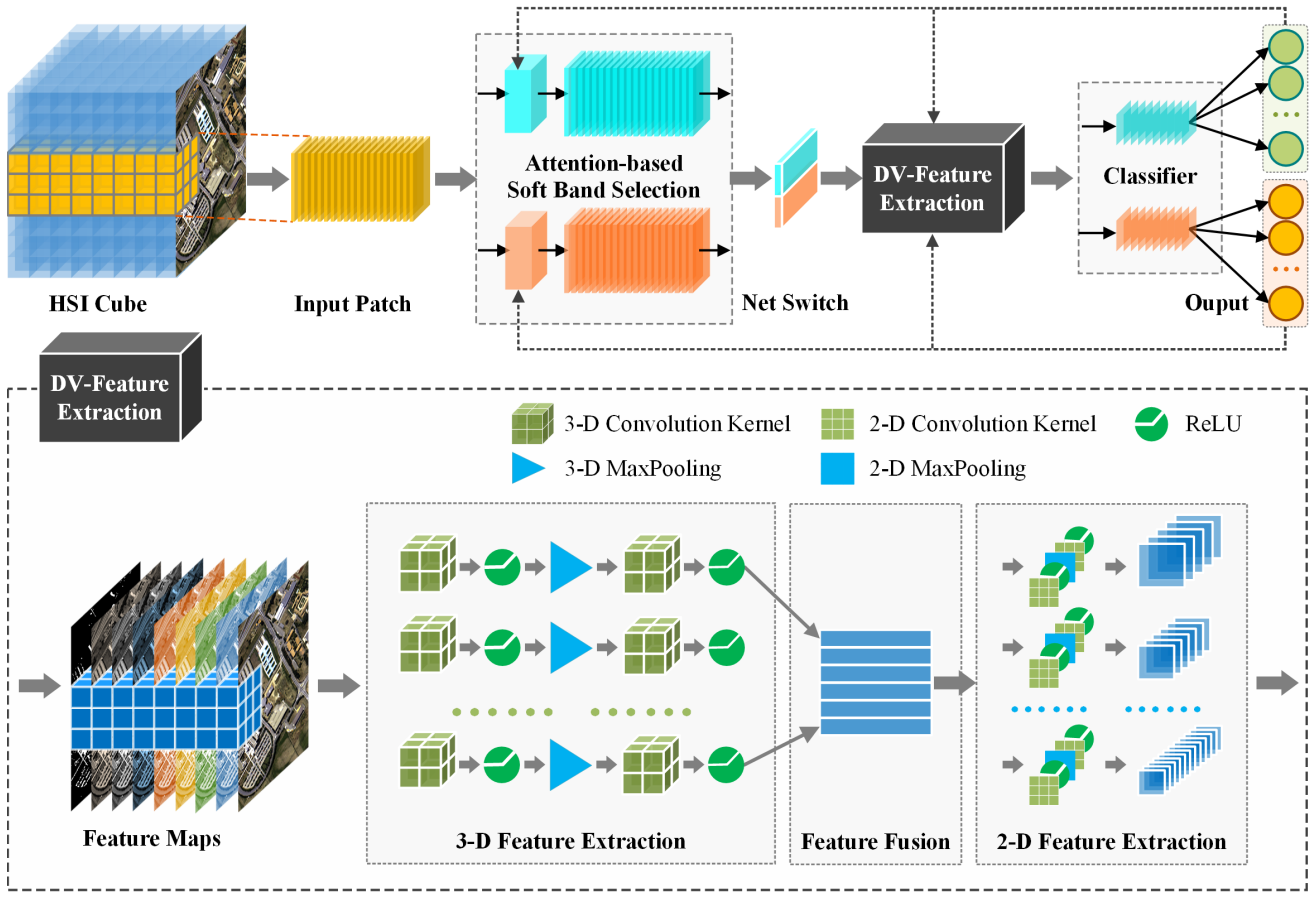


Fig. 3. Overview of ASNN architecture.

as follows:

$$\begin{aligned} P_j &= P_i + \Delta P \\ &= P_i - \eta \nabla P. \end{aligned} \tag{13}$$

There are two types of parameters in process P, and they are filter parameters in convolution kernels and biases in functions with η being the learning rate. The specific updating formulas are given by

$$\begin{aligned} \omega_j &= \omega_i - \eta \nabla \omega \\ &= \omega_i - \eta \frac{\partial E(\omega, b)}{\partial \omega} \end{aligned} \tag{14}$$

$$\begin{aligned} b_j &= b_i - \eta \nabla b \\ &= b_i - \eta \frac{\partial E(\omega, b)}{\partial b} \end{aligned} \tag{15}$$

where ω and b are the filter parameter and bias, respectively, and the loss function $E(\omega, b)$ can be calculated as

$$E(\omega, b) = -\sum_{k=1}^{N} y_k \ln(S_k) \tag{16}$$

where N is the total number of ground truth classes, and y_k and S_k are the ground truth label and classification output of the kth class, respectively. S_k can be calculated as

$$S_k = \frac{e^{a_k}}{\sum_{n=1}^{N} e^{a_n}} \tag{17}$$

where a_k and a_n are the kth and nth value in the feature vector generated by DVFE and taken as input by the classifier.

The learning process and symbiotic mechanism of ASNN are as follows. ASNN repetitively learns the sample distribution of ground truth labels and RWC labels from multilabel data set successively. The two classification processes share the same DVFE module, which means that they both contribute to the parameter update of DVFE by feedback. Afterward, well-trained DVFE can extract more accurate features for both processes. In other words, the two processes improve the accuracy of each other and the performance of the entire model within a symbiotic relationship. Besides, as a deep learning algorithm, ASNN is a data-driven model basically, and the features extracted during classification still depend on sample distribution and gradient effect. Nevertheless, the RWC labels are generated on the basis of corresponding physical bands at the specific wavelengths, which are informative and sensitive to RWC. In this situation, we can consider that DVFE integrates the physical–chemical features related to RWC, and ASNN is more explicable than pure data-driven models.

2) Soft Band Selection Based on Attention Mechanism: SBS is important for guiding different attention to the two classification tasks in ASNN. HSI data contain hundreds of bands, and each of them does not make equal contributions to each task. Some bands in low signal-to-noise ratio (SNR) may contain much noise, which is not conducive to tasks and may increase the computation. The purpose of

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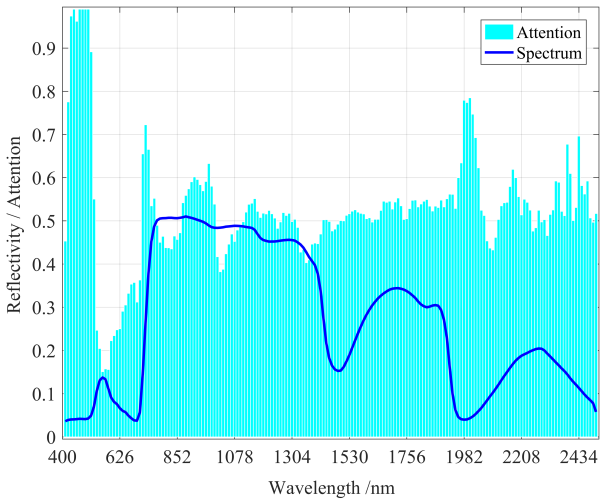


Fig. 4. SBS for spectral information.

band selection is to preserve the informative and relevant subset of bands and to reduce the redundancy or noise. Numerous bands' selection algorithms were proposed in the past few decades [60]. In terms of the algorithm improvement, ranking-based selection and searching-based selection algorithms made a significant progress. The clustering-based selection methods achieved remarkable improvement as well. Besides, some other methods such as sparsity-based methods, embedding learning-based methods, and hybrid scheme-based methods were more popular because of their practicality. In addition, several novel band selection methods were investigated based on deep neural networks [61]–[63] which aimed to select the most informative and discriminative bands for training and updating. Basically, band selection methods for HSI data exclusively select specific bands and discard other bands, which can be called hard band selection (HBS). Apparently, data loss caused by HBS implies that the data-driven deep models cannot extract any feature from the lost bands.

Accordingly, the SBS algorithm based on attention mechanism is proposed to deal with HSI data in ASNN. As opposed to HBS, SBS takes all weighted bands for feature extraction and updates weights by backpropagation algorithm. The SBS is a forward-propagated module, and the HSI bands input into the SBS are weighted and outputted to the next modules in ASNN. The weights of all bands are generated based on the attention mechanism, which is motivated by the human visual attention of cognitive activities [64]. The attention mechanism is responsible for assigning weights to spectral bands [65], [66], and each band weight in the attention mask represents the contribution to classification. Essentially, the goal of SBS is to obtain a weight matrix with the same dimensions of input HSI data. The exhibition of attention mask values and spectral bands is shown in Fig. 4. The two classification processes in ASNN both have corresponding attention masks of SBS, and the attention weights in each cluster both have two major characteristics. The first is that all the values of weights represent the contributions of corresponding bands for the classification task. The second is that the distribution of weights in each cluster establishes the relationship among bands, and it is able to reflect the specific attention of ASNN to the current classification task. Therefore, the attention-based SBS module in ASNN is able to pay more attention to informative bands and reduce the influence of trivial bands during feature extraction.

We focus on the importance of spectral attention rather than spatial attention, and a global attention mask $\mathbf{A}$ for all bands is generated, which is a 1-D vector initiated randomly. In practice, we always assign the same values in the range [0, 1] to initialize the mask $\mathbf{A}$. However, considering the deep structure of the classification model, the initialized $\mathbf{A}$ is usually multiplied by $(1 + \mathbf{A})$ to avoid gradient disappearance in deep layers. The function $f_{\mathrm{AS}}(\mathbf{X})$ abovementioned in (12) can be present as follows:

$$\begin{aligned}\mathbf{B} &= f_{\mathrm{AS}}(\mathbf{X}) \\ &= \mathbf{A}^{(\mathrm{tiled})} \cdot \mathbf{X}\end{aligned} \tag{18}$$

where $\mathbf{X}$ and $\mathbf{B}$ are the input patch and output patch of the SBS module, respectively. As $\mathbf{A}$ is an all-bands-attention vector of a spatial pixel in the 3-D patch $\mathbf{X}$, we need to tile it for all spatial pixels and take them as $\mathbf{A}^{(\mathrm{tiled})}$. Then, $\mathbf{A}^{(\mathrm{tiled})}$ is multiplied to the input patch $\mathbf{X}$ to produce the attention weighted data $\mathbf{B}$, which is taken as the input of the following modules for deep and abstract feature extraction. $\mathbf{B}$ has the same dimensions as $\mathbf{X}$. All the weights in the attention mask $\mathbf{A}$ can be trained and updated in the iteration of ASNN, and every negative weight value is reset to 0 after each iteration in training.

In addition, there are some differences between attention-based SBS and other analogous algorithms, such as SE block [67] or weighting-based methods [68]. The first and foremost difference is that the method is designed for ASNN to produce multiple attention masks for different classification tasks. Each of the attention masks focuses on a specific job and guides the following feature extraction. The second difference is that we focus on spectral information instead of spatial information without the global spatial average pooling in the SE block. The third difference is that we set the SBS module for global attention of all bands before feature extraction, and ASNN automatically learns different attention masks based on multilabel data to maintain the completeness of biophysical or biochemical characteristics in bands.

3) Dimensionality-Varied Feature Extraction: As we mentioned above, the designed DVFE is the core part of HSI for the synergy of spectral–spatial features extraction and even the reduction of computational cost. In many CNN-based models, because of the low spatial resolution and high spectral resolution of HSI, spectral feature extraction stops early when the spatial dimensionality is not able to be reduced any more. Moreover, abundant spectral features cannot be further extracted for refined classification. Therefore, we design the DVFE module on the basis of HSI data structural characteristics, which is capable of gathering global information after feature extraction by 3-D operations and extracting fused spectral–spatial–RWC features by 2-D operations in the deeper layers. The global information based on attention weights can focus on the interested features and discard the noisy feature in deeper feature extraction processes. Compared with

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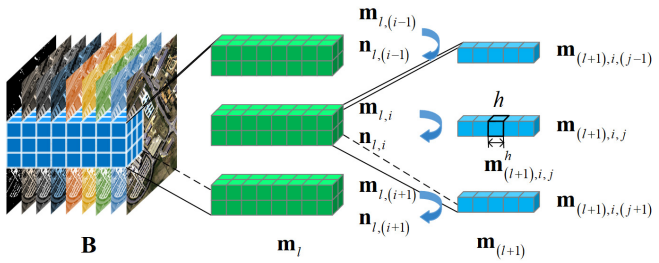


Fig. 5. 3-D feature extraction.

the state-of-the-art 3-D CNNs, the DVFE module with the transformation of the feature maps requires fewer parameters and less computation when the numbers of kernel and layer are unanimous. Besides, the DVFE module in ASNN is also updated by both supervised tasks, which result in the abundant features of RWC and spatial–spectral features. In general, the DVFE module is not only able to extract spectral–spatial features with the deeper network and less computation but also compatible with the SBS and RWCR processes in ASNN.

Most of the pixels in a certain spatial region of HSI data represent the same class of ground objects. It was demonstrated that spectral–spatial classification methods outperform spectral-only classification methods [69]. For this reason, DVFE takes the patch $\mathbf{B}$ as the input of $f_{\text{DEFV}}(\cdot)$ in (12), extracting features by 3-D and 2-D operations.

The process of 3-D feature extraction is shown in Fig. 5. We regard a group of sequential operations as a feature extraction layer, which contains a convolution operation at least. The first 3-D convolution, ReLU, and maxpooling operations compose the lth feature extraction layer, which generate numerous feature maps $\mathbf{m}_l$. Similarly, the second 3-D convolution and activation operations compose the $(l+1)$th layer which generates feature maps $\mathbf{m}_{(l+1)}$. Considering the input data $\mathbf{B}$ as the input layer data $\mathbf{m}_{(l-1)}$, the feature extraction of the lth layer can be formulated as follows:

$$\mathbf{m}_{l,i} = f_{\text{ReLU}}(\text{conv}_{3\text{-D}}(\mathbf{m}_{(l-1)}, \mathbf{k}_{l,i}) + \mathbf{b}_{l,i}) \tag{19}$$

$$\mathbf{n}_{l,i} = \text{maxpooling}(\mathbf{m}_{l,i}) \tag{20}$$

where $\mathbf{k}_{l,i}$ indicates the ith convolution kernel of the lth layer with $\mathbf{b}_{l,i}$ being the corresponding bias. $\text{conv}_{3\text{-D}}(\cdot)$ and $f_{\text{ReLU}}(\cdot)$ represent convolution operation and activation function, respectively. The calculation process of $\text{conv}_{3\text{-D}}(\cdot)$ can be reformulated as

$$\text{output}[m'] = \sum_{x=1}^{X}\sum_{y=1}^{Y}\sum_{z=1}^{Z} \text{input}[m^{g(x),g(y),g(z)}]k^{x,y,z} \tag{21}$$

$$g(x) = x + s(h-1) \tag{22}$$

where $k^{x,y,z}$ indicates one of the elements in 3-D convolution kernel $\mathbf{k}_{l,i}$ with (x, y, z) being the coordinate of element position. Similarly, $m^{g(x),g(y),g(z)}$ is one of the elements in the input feature map $\mathbf{m}_{(l-1)}$ with $(g(x), g(y), g(z))$ being the coordinate of element position. X, Y, and Z represent the length, width, and height of the 3-D kernels, respectively. m' is the outputted feature map after 3-D convolution. s is the stride of the convolution kernel with h being the moving steps of the convolution kernel. Moreover, the activation function is an unsaturated function called rectified linear units (ReLUs), which is faster than other saturation methods. As the following step of (19), maxpooling operation in (20) takes $\mathbf{m}_{l,i}$ as the input to reduce its dimensionality and generates $\mathbf{n}_{l,i}$. The feature extraction of the $(l+1)$th layer is similar to the lth layer without maxpooling operation.

The process of 2-D feature extraction is basically the typical operations for 2-D images. On the contrary to the 3-D feature extraction process, 2-D process takes the 2-D feature map fused by the 1-D vectors as input, which reduces lots of computation. Besides, 2-D process is capable of establishing the relationship between different 1-D feature vectors and extracting more abstract features. The extraction layers in 2-D process are composed of 2-D convolution, maxpooling, and some other operations. The specific description is elaborated in the experimental setting section. We consider each operation as a layer in this process. The convolution is given by

$$\mathbf{m}_{l,i} = f_{\text{ReLU}}(\text{conv}_{2\text{-D}}(\mathbf{m}_{(l-1)}, \mathbf{k}_{l,i}) + \mathbf{b}_{l,i}) \tag{23}$$

where l and i still represent the indices of layer and kernel, respectively, and $\text{conv}_{2\text{-D}}(\cdot)$ is the 2-D convolution operation. The calculation process of $\text{conv}_{2\text{-D}}(\cdot)$ can be reformulated as

$$\text{output}[m'] = \sum_{x=1}^{X}\sum_{y=1}^{Y} \text{input}[m^{g(x),g(y)}]k^{x,y}. \tag{24}$$

Other symbols in (23) and (24) basically remain the same meanings as (19)–(22) but all are mapped to 2-D images. The maxpooling function is basically the same as that in (20).

As the following step of 3-D process, it is noteworthy that the input of the 2-D process is a fused feature map, and the parameters in 2-D kernels are far fewer than that in 3-D kernels. With the simplified input and kernels, 2-D process can not only accelerate the computational speed substantially but also improve the capacity for increasing kernels and constructing a deeper network. Most importantly, the variation of feature maps in DVFE enhances the synergy of spectral–spatial feature extraction and makes it possible to further extract fused features in the deeper layers. Hence, DVFE is able to reinforce the synergistic representation and expression of spectral–spatial features for refined classification without computation increase.

III. Experiments and Discussion

In this article, the experiments are divided into two parts: 1) data augmentation based on RWCR inversion for ASNN and 2) HSI data refined classification by ASNN. In the second part, some ablation and comparison experiments were carried out to analyze the performance of ASNN. We used a hyperspectral data set of leaves acquired in the laboratory to verify the effectiveness of RES. Two popular real HSI data were used for the effectiveness validation of critical modules in ASNN and the superiority evaluation of ASNN in experiments.

A. Hyperspectral Data Sets

1) Leaves' Data Set: Leaves' data set was collected in the laboratory by the HeadSpecIII imaging system, which has

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Fig. 6. Leaves samples. (a) No. 1 sample with highest level water content. (b) No. 2 sample with the second-level and the fourth-level water content. (c) No. 3 sample with the third-level water content. (d) No.4 sample with the lowest level water content.

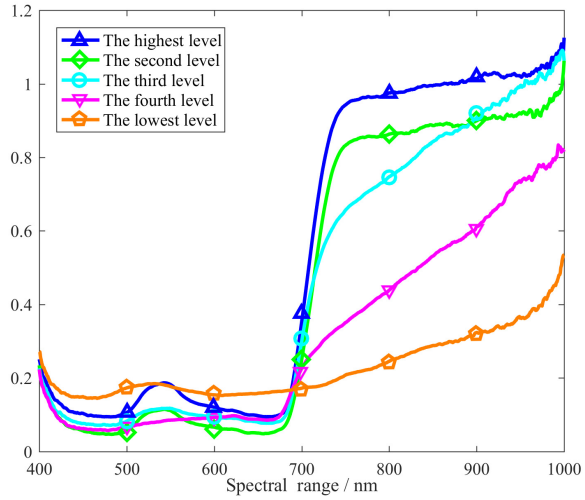


Fig. 7. Spectral curves of leaves with different water content levels.

a 380–1000 nm spectral range, and the effective range is 400–800 nm. HeadSpecIII also has 314 spectral bands, and the spectral resolution is smaller than 7 nm. The leaves are shown in Fig. 6, and the spectral curves of leaves are shown in Fig. 7.

As shown in Fig. 6, there are four leaves collected from the identical plant and presented according to their water content. We select five different areas to collect hyperspectral data from four leaves. The first area is selected from the No. 1 sample from Fig. 6(a) which is a fresh leaf with the highest-level water content. Then, we select two different leaf areas: the second-level (the main part of the No. 2 sample) and the fourth-level (the bottom right part of the No. 2 sample) water content areas from Fig. 6(b). We also select another area from the No. 3 sample with the third-level water content. Finally, we select the last area from the No. 4 sample, which is a plastic simulated leaf with the lowest water content. Fig. 7 shows the spectral differences of leaves with different water content in the wavelength range of 400–1000 nm. It is obvious that the differences mainly concentrated in the range of 650–760 nm, and the slope of each curve increases with the water content. This diagnostic signature can be used to distinguish the RWC of ground objects and improve the performance of refined classification.

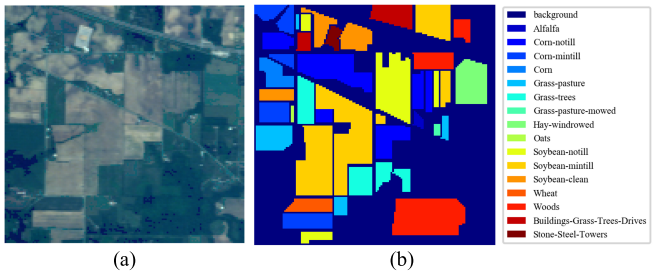


Fig. 8. Indian Pines data set composite. (a) False-color image. (b) Ground truth image.

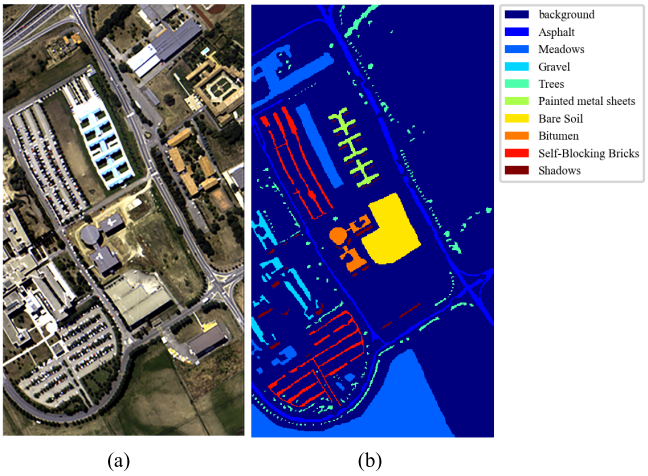


Fig. 9. Pavia University scene data set composite. (a) False-color image. (b) Ground truth image.

2) Indian Pines' Data Set: The Indian pines' data set was gathered by the AVIRIS sensor in 1992, and it covers the agricultural fields of Indian pines' test site in northwestern Indiana, USA. The image of the data set contains 145×145 pixels with the 20 meters per pixel (mpp) spatial resolution. There are 220 spectral bands in the wavelength range of 400–2500 nm, and 20 of them were removed because of atmospheric water absorption or null bands. Based on the ground-truth matrix, a total of 16 mutually exclusive ground truth classes in the data set are shown in Fig. 8.

3) Pavia University Scene Data Set: The Pavia University scene data set was acquired by the ROSIS sensor in 2001, and it covers an urban environment of the campus of Pavia University in Northern Italy. This data set contains 610×340 pixels with 1.3 mpp spatial resolution in this image. There are 103 spectral bands in the wavelength range of 430–860 nm with some noise-corrupted bands removed. Based on the ground-truth matrix, a total of nine mutually exclusive ground truth classes in the data set are shown in Fig. 9.

B. Experimental Settings

The small samples in HSI data sets have been reduplicated several times to keep sample balance during training. We randomly select 15% of the labeled data for training, following the same percentages proposed in [70]. The detailed figures of classes for experiments are shown in Tables I and II.

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TABLE I

TRAINING NUMBERS, TESTING NUMBERS, AND REDUPLICATION TIMES OF GROUND-TRUTH CLASSES IN INDIAN PINES

No.	Class	Training	Testing	Times
1	Alfalfa	7	46	50
2	Corn-notill	214	1428	1
3	Corn-mintill	125	830	1
4	Corn	36	237	10
5	Grass-pasture	72	483	5
6	Grass-trees	110	730	1
7	Grass-pasture-mowed	4	28	80
8	Hay-windrowed	72	478	1
9	Oats	3	20	100
10	Soybean-notill	146	972	1
11	Soybean-mintill	368	2455	1
12	Soybean-clean	89	593	1
13	Wheat	31	205	10
14	Woods	190	1265	1
15	Buildings-Grass-Trees-Drives	58	386	5
16	Stone-Steel-Towers	14	93	20
	Total Samples	1537	10249	25916

TABLE II

TRAINING NUMBERS, TESTING NUMBERS, AND REDUPLICATION TIMES OF GROUND-TRUTH CLASSES IN PAVIA UNIVERSITY SCENE

No.	Class	Training	Testing	Times
1	Asphalt	995	6631	2
2	Meadows	2797	18649	1
3	Gravel	315	2099	9
4	Trees	460	3064	6
5	Painted metal sheets	202	1345	14
6	Bare Soil	754	5029	4
7	Bitumen	200	1330	14
8	Self-Blocking Bricks	552	3682	5
9	Shadows	142	947	20
	Total Samples	6416	42776	164102

The HSI data argumentation based on the RWCR inversion model and classification-based ASNN share the same experimental environment. The experiments were implemented based on the Tensorflow1.12 platform in Windows OS and conducted using NVIDIA GTX 1080Ti GPU and Intel Core i7-8700K at 3.4 GHz with 64 GB of RAM. In addition, the Cuda version and cuDNN version were 9.0 and 7.5, respectively.

ASNN randomly takes 50 and 100 training samples from Indian Pines data set and Pavia University scene data set, respectively, as the input mini-batches. The model is repetitively trained with softmax loss, and the training epochs are set as 5000 to ensure convergence. All data are shuffled after every epoch to avoid recurring patterns in backpropagation and to make the process smoother. Some feature extraction parameters of ASNN are shown in Table III, and the other fundamental hyperparameters of ASNN are set as follows. m and n in Table III represent the adjustable window size of the convolution kernel for input data, and other window sizes are basically set as 3×3. We empirically set the learning rate as 1×10^{-4} to control the update magnitude of weights in each epoch. It is able to achieve a relative equilibrium between converge speed and loss reduction, avoiding diverging or oscillating around the local minimum. All weights in convolution layers are initialized randomly, and the optimization algorithm

TABLE III

PARTIAL PARAMETER SETTINGS OF ASNN

Module	Layer	Kernel Size	Kernel Number
3-D Feature Extraction	3-D Convolution	$m \times m \times n$	10
	Max pooling	2×2×2	—
	3-D Convolution	3×3×3	20
2-D Feature Extraction	2-D Convolution	3×3	24
	Max pooling	2×2	—
	2-D Convolution	3×3	36
	2-D Convolution	3×3	48
	2-D Convolution	3×3	60
	2-D Convolution	3×3	60
Fully Connection	Fully Connection	2400×1024	1

of ASNN is set as Adam optimizer. In addition, to alleviate the randomness of results, we carried out the independent classification experiments for ten Monte Carlo runs to obtain the average and standard deviation of metrics.

In order to evaluate the performance of ASNN, we select some metrics to assess the classification results. First, we adopt three quantitative metrics to assess the interclass classification performance: OA, average accuracy (AA), and kappa coefficient (κ). Second, we use the confusion matrix to analyze the misclassified categories for the assessment of intraclass classification performance. Here, OA represents the classification capability of the model for all samples. AA represents the average classification performance for all classes. Kappa coefficient represents the strong agreement based on statistical measurement between classification results and ground truths. The confusion matrix is able to show the misclassification among all categories.

C. Experiments and Discussion

The experiments and discussion can be divided into three parts according to experimental settings. In the first part, we conducted RWCR inversion experiments based on prior physical knowledge for ASNN data augmentation. In the second part, we analyzed the influences of the spatial size of ASNN and carried out a set of ablation studies to test the improvements of RWCR augmentation and SBS for ASNN. In the final part, we evaluated the refined classification performance of ASNN. In specific, we analyzed the misclassified classes in confusion matrix and assessed the intraclass improvement. We also carried out comparison experiments with state-of-the-art methods for interclass improvement analysis. A total of five different state-of-the-art classification approaches (SVM, RF, MLP, 2-D CNN, and 3-D CNN) in [70] and [71] have been selected to compare with ASNN, and all experimental settings stay the same for a fair comparison.

1) RWCR Inversion for Data Augmentation of ASNN: This part of the experiments consists of two steps. First, we utilize leaves' data set to test the effectiveness and superiority of RES for RWCR. Second, we implement the RES calculation and adaptive grading algorithm for the generation of RWC labels of two other HSI data sets, such as Indian Pines and Pavia

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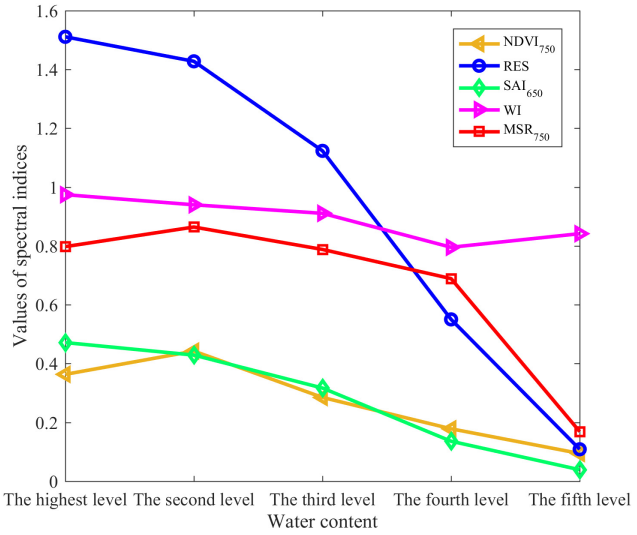


Fig. 10. Variation curves of spectral index values with water content.

University scene. Then, the multilabel data sets are constructed and taken by ASNN as the augmented input for training and refined classification.

Fig. 10 shows the variations of spectral index values with the water content of leaves. The variations of $NDVI_{750}$, MSR_{750}, and WI are irregular, which are not applicable to RWCR. SAI_{650} and RES both can characterize the variations of RWC. Especially, RES is preferable to represent the variations because the RES values decrease significantly with the water content of leaves. Therefore, RES is selected as the most effective spectral index to retrieve the RWC of HSI data.

All RES values of HSI data are calculated and graded into different levels by adaptive grading algorithm. Each level corresponds to an RWC label. There are 16 classes and 9 classes of ground objects in Indian pines and Pavia University scene, and we create five RWC labels (the highest level, the higher level, the medium level, the lower level, and the lowest level) and three RWC labels (the highest level, the medium level, and the lowest level) for grading, respectively. The RWC labeling results of Indian Pines and Pavia University scene are shown in Tables IV and V, respectively. All RWC labels are attached to the original HSI data for multilabel data sets construction.

2) Ablation Study of ASNN: In this part of the experiments, we test the effects of the spatial size of input data and two critical parts of ASNN. Spatial size is an important factor affecting spatial information extraction. We empirically select six different spatial sizes, which are 1×1 (spectral-only), 7×7, 9×9, 11×11, 13×13, and 15×15 to test the effects for classification. The optimal size is taken as the default hyperparameters of ASNN for the following experiments. Moreover, we also conduct a set of ablation experiments to verify the improvements of the SBS module and data augmentation based on RWCR.

The classification results of ASNN with different spatial sizes are shown in Table VI. Obviously, the classification accuracy varies with the spatial size, and the variation trend on each data set is basically consistent with that on the others. The results suggest that spectral-only classification (1×1), which discards spatial information, achieves the lowest accuracy. An 11×11 is the most suitable spatial size for ASNN. The classification maps of ASNN with 11×11 spatial size on Indian Pines and Pavia University scene are shown in Figs. 11 and 12, respectively. Especially, it is possible to observe that the classification map of Pavia University scene is similar to the ground truth map, which demonstrates the high classification accuracy of ASNN.

The classification results of the ablation experiments are shown in Table VII. There are four models (DVFE, SBS, and RWCR) in Table VII, and all of them are implemented based on ASNN architecture. The SE-Block in [31] is added

TABLE IV
RWC LABELING RESULTS OF INDIAN PINES

No.	Ground-truth type	RWC label
1	Alfalfa	High
2	Corn-notill	Lower
3	Corn-mintill	Lower
4	Corn	Lower
5	Grass-pasture	Higher
6	Grass-trees	Higher
7	Grass-pasture-mowed	High
8	Hay-windrowed	Higher
9	Oats	Higher
10	Soybean-notill	Lower
11	Soybean-mintill	Lower
12	Soybean-clean	Lower
13	Wheat	Lower
14	Woods	Medium
15	Buildings-Grass-Trees-Drives	Medium
16	Stone-Steel-Towers	Low

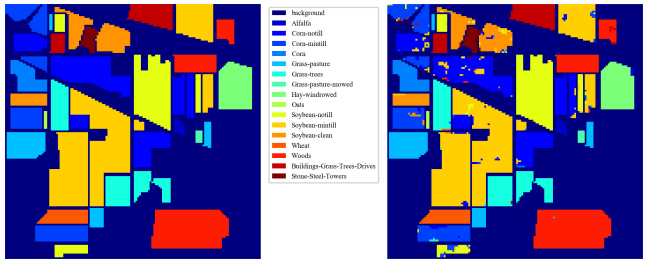
TABLE V
RWC LABELING RESULTS OF PAVIA UNIVERSITY SCENE

No.	Ground-truth type	RWC label
1	Asphalt	Low
2	Meadows	High
3	Gravel	Low
4	Trees	High
5	Painted metal sheets	Low
6	Bare Soil	Medium
7	Bitumen	Low
8	Self-Blocking Bricks	Low
9	Shadows	Medium

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT SPATIAL SIZES OF ASNN

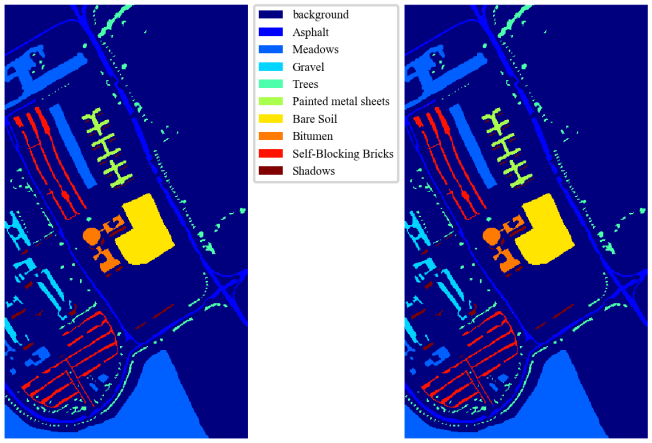
Data set	Spatial size	OA	AA	κ
Indian Pines	1×1	90.26%	86.24%	0.8947
	7×7	98.06%	97.49%	0.979
	9×9	98.49%	97.76%	0.9837
	11×11	**99.23%**	**98.82%**	**0.9917**
	13×13	99.17%	98.68%	0.9911
	15×15	98.83%	98.08%	0.9874
Pavia University	1×1	89.85%	87.00%	0.8648
	7×7	99.21%	98.64%	0.9895
	9×9	99.56%	99.44%	0.9941
	11×11	**99.96%**	**99.96%**	**0.998**
	13×13	99.63%	99.44%	0.9951
	15×15	99.69%	99.47%	0.9958

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(a) (b)

Fig. 11. Classification results of ASNN with 11×11 spatial size on Indian Pines. (a) Ground truth map. (b) Classification map.



(a) (b)

Fig. 12. Classification results of ASNN with 11×11 spatial size on Pavia University scene. (a) Ground truth map. (b) Classification map.

TABLE VII
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS

Model	Indian Pines			Pavia University scene		
	OA	AA	κ	OA	AA	κ
DVFE Only	98.06%	97.13%	0.9791	99.50%	99.20%	0.9934
SBS+DVFE	98.90%	98.29%	0.9882	99.73%	99.57%	0.9964
SE-Block+DVFE	98.39%	97.92%	0.9816	99.63%	99.51%	0.9948
RWCR+DVFE	98.82%	98.13%	0.9873	99.74%	99.56%	0.9965
ASNN	**99.23%**	**98.82%**	**0.9917**	**99.96%**	**99.96%**	**0.9998**

for comparison, which is a similar module with the SBS module. The DVFE only is the implementation of ASNN without the SBS module, and it takes the original data as input without RWCR augmentation. The model SBS+DEVFE is the implementation of ASNN with the SBS module, which takes original data as input too. The model SE-Block + DVFE is similar to the SBS+DVFE model. The model RWCR+DVFE is the implementation of ASNN without the SBS module, which takes augmented multilabel data as input. The last model is a complete ASNN, which contains the SBS module and takes augmented multilabel HSI data as input.

The results reported in Table VII exhibit similar behavior on both data sets. It is obvious that the two critical parts, SBS module and data augmentation based on RWCR, both can improve the classification performance significantly, and the improvement of SBS is more than the SE-Block. Moreover,

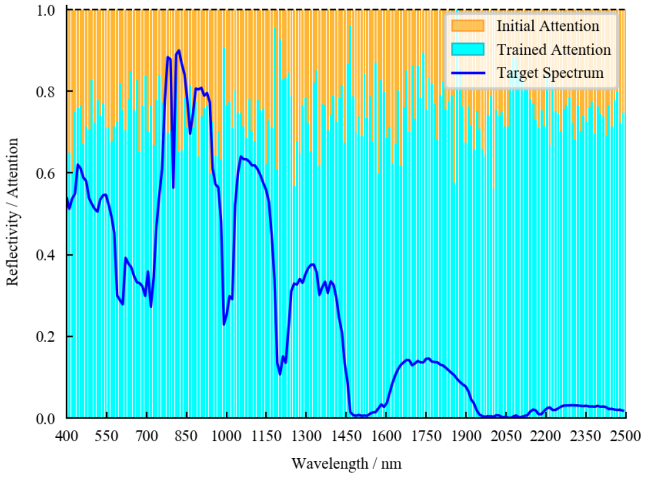


Fig. 13. SBS of ASNN on Indian Pines.

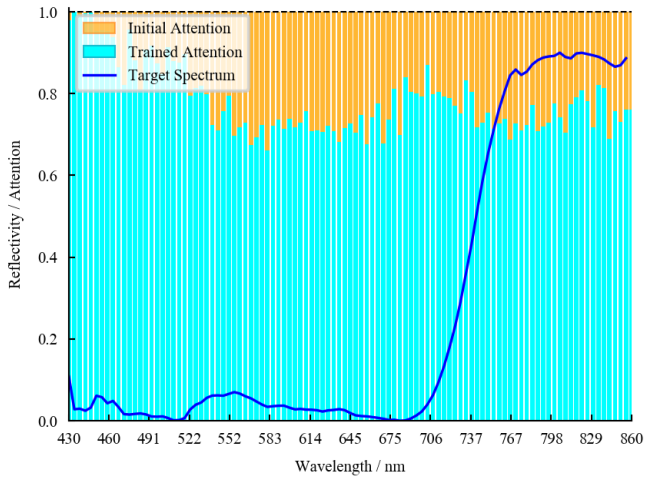


Fig. 14. SBS of ASNN on Pavia University scene.

ASNN with the critical parts achieves the best performance. For illustrative purpose, the attention mask of SBS on both data sets is shown in Figs. 13 and 14, respectively. Before model training, we set weights of all bands as ones, which means that all bands are initialized with full attention (illustrated as the orange bars) for training. After iterative training, the attention of every band is different from that of each other (illustrated as the cyan bar), which represents the different contributions of bands to classification. For the quantitative analysis, the OA improvements of ASNN over the DVFE only, SBS+DVFE, and RWCR+DVFE models are +1.17%, 0.33%, and 0.41%, respectively, on Indian Pines data set, and +0.46%, 0.23%, and 0.22%, respectively, on Pavia University scene data set (as the AA and Kappa coefficient also suggest). Therefore, the effectiveness of SBS and RWCR data augmentation in ASNN is demonstrated in the quantitative and qualitative ways. The proposed ASNN, which is combined with the two effective parts, can achieve a better classification performance.

3) Performance Analysis of ASNN Refined Classification: This part of the experiments tests the refined classification performance and verifies the superiority of ASNN. Specifically, we divide the experiments into two groups. The first group

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TABLE VIII
CONFUSION MATRIX OF DVFE MODULE ON INDIAN PINES

	C-1	C-2	C-3	C-4	C-5	C-6	C-7	C-8	C-9	C-10	C-11	C-12	C-13	C-14	C-15	C-16	Accuracy
C-1	2300	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	100.00%
C-2	**0**	**1300**	**45**	**10**	**4**	**0**	**3**	**2**	**1**	**15**	**30**	**10**	**0**	**0**	**8**	**0**	**91.04%**
C-3	**0**	**25**	**737**	**5**	**5**	**2**	**2**	**1**	**1**	**27**	**11**	**11**	**0**	**3**	**0**	**0**	**88.80%**
C-4	0	0	0	2370	0	0	0	0	0	0	0	0	0	0	0	0	100.00%
C-5	0	0	0	0	2400	0	0	0	0	0	0	0	0	10	5	0	99.38%
C-6	**0**	**0**	**3**	**0**	**12**	**694**	**0**	**0**	**6**	**0**	**0**	**0**	**0**	**13**	**2**	**0**	**95.07%**
C-7	0	0	0	0	0	0	2240	0	0	0	0	0	0	0	0	0	100.00%
C-8	0	0	0	0	1	0	0	477	0	0	0	0	0	0	0	0	99.79%
C-9	0	0	0	0	0	0	0	0	2000	0	0	0	0	0	0	0	100.00%
C-10	0	0	10	2	2	3	0	0	0	940	12	2	0	1	0	0	96.71%
C-11	0	17	22	0	0	4	0	3	0	39	2350	12	0	4	4	0	95.72%
C-12	**0**	**13**	**6**	**23**	**0**	**1**	**0**	**1**	**0**	**0**	**7**	**534**	**1**	**0**	**0**	**7**	**90.05%**
C-13	0	0	0	0	0	0	0	0	0	0	0	0	2050	0	0	0	100.00%
C-14	0	0	0	0	6	1	0	0	0	0	0	0	0	1237	21	0	97.79%
C-15	0	0	0	0	0	0	0	0	0	0	0	0	0	5	1925	0	99.74%
C-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1860	100.00%

C-1 : Alfalfa C-2 : Corn-notill C-3 : Corn-mintill C-4 : Corn
C-5 : Grass-pasture C-6 : Grass-trees C-7 : Grass-pasture-mowed C-8 : Hay-windrowed
C-9 : Oats C-10: Soybean-notill C-11: Soybean-mintill C-12: Soybean-clean
C-13: Wheat C-14: Woods C-15: Buildings-Grass-Trees-Drives C-16: Stone-Steel-Towers

TABLE IX
CONFUSION MATRIX OF ASNN ON INDIAN PINES

	C-1	C-2	C-3	C-4	C-5	C-6	C-7	C-8	C-9	C-10	C-11	C-12	C-13	C-14	C-15	C-16	Accuracy
C-1	2300	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	100.00%
C-2	**0**	**1359**	**9**	**2**	**0**	**1**	**8**	**0**	**0**	**7**	**37**	**3**	**2**	**0**	**0**	**0**	**95.17%**
C-3	**0**	**3**	**786**	**2**	**0**	**3**	**1**	**0**	**0**	**18**	**13**	**1**	**0**	**0**	**3**	**0**	**94.70%**
C-4	0	0	0	2370	0	0	0	0	0	0	0	0	0	0	0	0	100.00%
C-5	0	0	0	0	2415	0	0	0	0	0	0	0	0	0	0	0	100.00%
C-6	**0**	**2**	**0**	**0**	**0**	**728**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**99.73%**
C-7	0	0	0	0	0	0	2240	0	0	0	0	0	0	0	0	0	100.00%
C-8	0	0	0	0	0	0	0	478	0	0	0	0	0	0	0	0	100.00%
C-9	0	0	0	0	0	0	0	0	2000	0	0	0	0	0	0	0	100.00%
C-10	0	0	0	0	0	0	0	0	0	964	4	0	0	1	3	0	99.18%
C-11	0	2	3	2	0	0	0	0	0	10	2436	0	0	2	0	0	99.23%
C-12	**0**	**1**	**4**	**13**	**1**	**0**	**0**	**0**	**0**	**3**	**3**	**563**	**0**	**0**	**0**	**5**	**94.94%**
C-13	0	0	0	0	0	0	0	0	0	0	0	0	2050	0	0	0	100.00%
C-14	0	0	1	0	7	2	0	0	0	3	0	0	0	1240	12	0	98.02%
C-15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1930	0	100.00%
C-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1860	100.00%

C-1 : Alfalfa C-2 : Corn-notill C-3 : Corn-mintill C-4 : Corn
C-5 : Grass-pasture C-6 : Grass-trees C-7 : Grass-pasture-mowed C-8 : Hay-windrowed
C-9 : Oats C-10: Soybean-notill C-11: Soybean-mintill C-12: Soybean-clean
C-13: Wheat C-14: Woods C-15: Buildings-Grass-Trees-Drives C-16: Stone-Steel-Towers

is the evaluation of intraclass and interclass classification performance of ASNN by the confusion matrix. The second group assesses the superiority of ASNN in comparison with five state-of-the-art classifiers.

We assume that the classes with the same RWC labels or containing the same ground objects belong to the same category. For instance, ground objects with high-level water content or containing corns belong to the high-level or corns' category. In order to evaluate the intraclass and interclass classification performance of the same categories and the entire HSI data, we analyze the confusion matrices of classification results. The similar behavior is observed in ten independent results on both data sets, and one of them is randomly selected for the presentation. Tables VIII–XI show the comparison results of DVFE only and ASNN models on Indian Pines data set and Pavia University scene data set, respectively.

As shown in Tables VIII and IX, the classification accuracy of most classes on Indian Pines data set has been improved, and there are four classes with the improvements exceeding +4%. In particular, the accuracy improvements of ASNN over DVFE only are +4.13%, +5.90%, +4.66%, and +4.89% on corn-notill (C-2), corn-mintill (C-3), grass-trees (C-6), and soybean-clean (C-12), respectively. The number of corn-notill (C-2) misclassified into corn-mintill (C-3) is reduced by 36, which decreases the misclassification accuracy by 80%. Besides, the misclassification accuracy of corn-mintill (C-3) into corn-notill (C-2) is decreased by 88%, and the misclassification accuracy of Soybean-clean (C-12) into corn-notill (C-2) and corn (C-4) is decreased by 92.31% and 43.48%, respectively. Similarly, the discrimination of objects in the same category (with the same RWC labels) is increased due to RWCR data augmentation. The misclassification

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TABLE X
CONFUSION MATRIX OF DVFE ONLY ON PAVIA UNIVERSITY SCENE

	Asphalt	Meadows	Gravel	Trees	Painted metal sheets	Bare Soil	Bitumen	Self-Blocking Bricks	Shadows	Accuracy
Asphalt	13214	0	10	0	0	2	0	34	2	99.64%
Meadows	0	18649	0	0	0	0	0	0	0	100.00%
Gravel	**18**	**0**	**18630**	**0**	**0**	**0**	**9**	**225**	**9**	**98.62%**
Trees	**18**	**264**	**0**	**18042**	**0**	**6**	**0**	**12**	**42**	**98.14%**
Painted metal sheets	0	0	0	0	18830	0	0	0	0	100.00%
Bare Soil	0	92	0	0	0	20004	0	20	0	99.44%
Bitumen	140	0	0	0	0	0	18480	0	0	99.25%
Self-Blocking Bricks	**130**	**0**	**120**	**30**	**0**	**0**	**0**	**18120**	**10**	**98.42%**
Shadows	20	60	20	40	0	0	0	0	18800	99.26%

TABLE XI
CONFUSION MATRIX OF ASNN ON PAVIA UNIVERSITY SCENE

	Asphalt	Meadows	Gravel	Trees	Painted metal sheets	Bare Soil	Bitumen	Self-Blocking Bricks	Shadows	Accuracy
Asphalt	13244	0	0	0	0	0	2	16	0	99.86%
Meadows	0	18648	0	1	0	0	0	0	0	99.99%
Gravel	**0**	**0**	**18891**	**0**	**0**	**0**	**0**	**0**	**0**	**100.00%**
Trees	**6**	**0**	**0**	**18378**	**0**	**0**	**0**	**0**	**0**	**99.97%**
Painted metal sheets	0	0	0	0	18830	0	0	0	0	100.00%
Bare Soil	0	0	0	0	0	20116	0	0	0	100.00%
Bitumen	0	0	0	0	0	0	18620	0	0	100.00%
Self-Blocking Bricks	**25**	**0**	**10**	**0**	**0**	**0**	**0**	**18375**	**0**	**99.81%**
Shadows	0	0	0	0	0	0	0	0	18940	100.00%

accuracy of grass-trees (C-6, higher-level RWC) into grass-pasture (C-5, higher-level RWC), woods (C-14, medium-level RWC), and corn-mintll (C-2, lower-level RWC) has also been decreased in a different extent. The similar behavior is shown in Tables X and XI, which report the classification confusion matrices on Pavia University data set. As shown in Tables X and XI, there are three classes with the improvements of accuracy exceeding +1%. In particular, the accuracy improvements of ASNN over DVFE only are +1.38%, +1.83%, and +1.39% on gravel, trees, and self-blocking bricks, respectively. The number of gravel misclassified into self-blocking bricks is decreased by 225, which completely reduce the misclassification. Conversely, the misclassification accuracy of self-blocking bricks into gravel is decreased by 91.67%. Similarly, the numbers of trees (high-level RWC) misclassified into meadows (high-level RWC), shadows (medium-level RWC), and bare soil (low-level RWC) have been reduced to 0, which improves the accuracy in a large extent.

The results shown in Tables VII–XI suggest that the multilabel data augmented by RWCR inject the inherent biochemical signatures into ASNN and improve the discrimination of similar targets. The SBS modules based on attention mechanism leads ASNN to focus on subtle intraclass difference and to extract diagnostic features of targets. That is, ASNN has the capacity for implementing refined classification and improving intraclass and interclass classification accuracy simultaneously.

In order to verify the superiority of ASNN, three different experiments are carried out. First, we conduct a comparison experiment between the proposed ASNN and some state-of-the-art classification methods: SVM, RF, MLP, 2-D CNN, and 3-D CNN. SVM, RF, and MLP are spectral classifiers, and 2-D CNN is a spectral–spatial classifier which extracts spectral features and spatial features separately. The 3-D CNN takes the 3-D cube of HSI data as input and extracts spectral–spatial features simultaneously. The classification results on Indian Pines and Pavia University scene are shown in Tables XII and XIII. It is noticeable that experimental results on two data sets show that the proposed method ASNN achieves the best classification performance. The accuracy improvements of ASNN over the second-best classifier 3-D CNN are +1.42%, 4.75%, and +1.67% on Indian Pines and +0.68%, +1.15%, and +6.93 on Pavia University scene for OA, AA, and Kappa coefficient, respectively. In addition, ASNN achieves the best accuracy in most classes of Indian Pines and each class of Pavia University scene. The standard deviations of OA, AA, and Kappa coefficient in the results of ASNN are much smaller than that in the results of other methods, which indicates that the proposed method has a better stability in training. Besides, according to the computational times, it is noticeable that MLP achieves the lowest computation time than other methods, followed by ASNN, 3-D CNN, and 2-D CNN. SVM and RF methods are the most time-consuming classifiers. Considering both classification accuracy and time-consuming, the proposed ASNN still achieves the best classification performance among those methods. Second, in order to verify the effect of training samples on the classification performance of ASNN, we carry out a group of experiments with different training ratios, which are 5%, 10%, 15%, 20%, and 25%. The classification results are shown in Table XIV. It is obvious that the OA, AA, and Kappa coefficient basically increase with the training ratio. Finally, we conduct the comparison experiments with the newest CNN-based models, 3-D-LWNet, and SSRN in [41] and [50]. We set different training samples for ASNN

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TABLE XII
CLASSIFICATION RESULTS OF ASNN ON INDIAN PINES (%)

No.	Ground-truth type	SVM	RF	MLP	2-D CNN	3-D CNN	ASNN
1	Alfalfa	68.04 ± 6.95	33.04 ± 7.45	62.39 ± 13.96	65.87 ± 10.34	89.13 ± 7.28	**100.00 ± 0.00**
2	Corn-notill	83.55 ± 1.31	66.68 ± 1.67	83.84 ± 2.46	81.04 ± 3.28	98.33 ± 0.71	95.07 ± 0.11
3	Corn-mintill	73.82 ± 1.44	56.20 ± 2.41	76.37 ± 5.03	79.07 ± 6.75	98.05 ± 1.40	96.21 ± 1.51
4	Corn	71.98 ± 3.86	41.10 ± 2.50	68.35 ± 6.12	82.70 ± 8.34	98.23 ± 0.62	**100.00 ± 0.00**
5	Grass-pasture	94.29 ± 0.97	87.12 ± 1.73	90.87 ± 2.09	69.25 ± 10.58	97.56 ± 2.84	**100.00 ± 0.00**
6	Grass-trees	97.32 ± 0.97	95.32 ± 1.79	96.95 ± 1.10	88.29 ± 5.51	98.93 ± 1.14	**99.11 ± 0.62**
7	Grass-pasture-mowed	88.21 ± 5.06	32.86 ± 12.66	78.21 ± 10.28	67.86 ± 25.65	83.57 ± 19.51	**100.00 ± 0.00**
8	Hay-windrowed	98.16 ± 0.75	98.49 ± 0.81	98.08 ± 0.90	96.26 ± 1.60	99.41 ± 0.61	**99.90 ± 0.11**
9	Oats	52.00 ± 8.43	13.00 ± 3.32	72.00 ± 8.12	67.00 ± 27.68	65.00 ± 21.68	**100.00 ± 0.00**
10	Soybean-notill	79.49 ± 2.76	69.95 ± 4.31	82.17 ± 5.41	68.82 ± 9.80	97.22 ± 0.31	**99.23 ± 0.02**
11	Soybean-mintill	86.83 ± 1.05	90.66 ± 1.18	83.66 ± 2.85	86.55 ± 3.14	98.12 ± 2.16	**98.90 ± 0.33**
12	Soybean-clean	83.41 ± 2.26	55.43 ± 4.80	75.89 ± 3.33	73.41 ± 6.07	93.09 ± 5.85	**94.45 ± 0.51**
13	Wheat	97.41 ± 2.99	93.32 ± 2.04	98.68 ± 0.54	94.54 ± 4.80	99.80 ± 039	**100.00 ± 0.00**
14	Woods	96.14 ± 0.97	96.45 ± 0.76	96.17 ± 1.02	96.24 ± 2.33	99.43 ± 0.33	98.18 ± 0.16
15	Buildings-Grass-Trees-Drives	67.31 ± 3.05	50.44 ± 2.44	67.80 ± 3.56	85.39 ± 7.71	96.58 ± 2.81	**100.00 ± 0.00**
16	Stone-Steel-Towers	92.47 ± 4.14	85.27 ± 3.37	88.71 ± 2.77	92.90 ± 3.97	93.12 ± 3.82	**100.00 ± 0.00**
	OA	86.24 ± 0.38	78.55 ± 0.68	85.27 ± 0.47	83.59 ± 0.88	97.81 ± 0.56	**99.23 ± 0.01**
	AA	83.15 ± 1.10	66.58 ± 0.93	82.51 ± 1.04	80.95 ± 1.55	94.10 ± 2.00	**98.82 ± 0.02**
	κ	84.27 ± 0.45	75.20 ± 0.81	83.20 ± 0.53	81.23 ± 1.04	97.50 ± 0.64	**99.17 ± 0.01**
	Time (s)	208.98 ± 1.70	1,301.68 ± 45.94	**7.31 ± 0.15**	56.45 ± 0.19	39.62 ± 0.67	38.69 ± 0.82

TABLE XIII
CLASSIFICATION RESULTS OF ASNN ON PAVIA UNIVERSITY SCENE (%)

No.	Ground-truth type	SVM	RF	MLP	2-D CNN	3-D CNN	ASNN
1	Asphalt	95.36 ± 0.30	93.52 ± 0.45	94.17 ± 1.73	93.43 ± 2.70	99.16 ± 0.25	**99.92 ± 0.06**
2	Meadows	98.25 ± 0.16	98.29 ± 0.18	98.06 ± 0.50	97.59 ± 0.88	99.77 ± 0.17	**99.98 ± 0.01**
3	Gravel	82.93 ± 0.917	5.56 ± 1.86	79.27 ± 7.04	89.96 ± 3.30	96.95 ± 1.78	**100.00 ± 0.00**
4	Trees	95.93 ± 0.70	91.68 ± 0.63	94.61 ± 2.58	94.16 ± 3.24	98.80 ± 0.69	**99.935 ± 0.04**
5	Painted metal sheets	99.46 ± 0.36	98.88 ± 0.49	99.63 ± 0.27	97.97 ± 2.69	99.90 ± 0.17	**100.00 ± 0.00**
6	Bare Soil	91.76 ± 0.60	74.54 ± 0.97	93.60 ± 1.70	89.62 ± 4.10	99.88 ± 0.12	**100.00 ± 0.00**
7	Bitumen	88.59 ± 0.65	81.01 ± 1.74	88.53 ± 3.47	80.20 ± 4.82	96.54 ± 1.41	**100.00 ± 0.00**
8	Self-Blocking Bricks	90.14 ± 0.54	90.70 ± 0.75	89.59 ± 4.56	96.05 ± 1.88	98.56 ± 0.78	**99.81 ± 0.00**
9	Shadows	99.97 ± 0.05	99.75 ± 0.26	99.63 ± 0.28	99.48 ± 0.27	99.79 ± 0.19	**100.00 ± 0.00**
	OA	95.20 ± 0.13	92.03 ± 0.21	94.82 ± 0.26	94.77 ± 0.72	99.28 ± 0.25	**99.96 ± 0.01**
	AA	93.60 ± 0.14	89.33 ± 0.33	93.01 ± 0.60	93.16 ± 1.23	98.81 ± 0.33	**99.96 ± 0.00**
	κ	93.63 ± 0.17	89.30 ± 0.28	93.13 ± 0.34	93.05 ± 0.97	99.04 ± 0.32	**99.98 ± 0.01**
	Time (s)	6,084.92 ± 55.64	6,188.75 ± 35.16	**29.10 ± 0.92**	172.29 ±0.71	140.09 ± 1.63	138.27 ± 1.31

TABLE XIV
CLASSIFICATION RESULTS OF ASNN WITH DIFFERENT TRAINING RATIOS

Training Ratio	Indian Pines			Pavia University scene		
	OA	AA	κ	OA	AA	κ
5%	98.87%	98.03%	0.9856	99.23%	99.17%	0.9926
10%	99.03%	98.26%	0.9879	99.82%	99.31%	0.9947
15%	99.23%	98.82%	0.9917	99.96%	99.96%	0.9998
20%	99.28%	98.89%	0.9903	99.96%	99.96%	0.9998
25%	**99.45%**	**99.02%**	**0.9924**	**99.97%**	**99.97%**	**0.9998**

TABLE XV
CLASSIFICATION RESULTS OF METHODS WITH DIFFERENT TRAINING SAMPLES

Data sets	Training Samples	3D-LWNet (%)	SSRN (%)	ASNN (%)
Indian Pines	2055	-	99.19±0.26	99.19±0.02
	1765	98.87±0.31	-	99.10±0.26
Pavia University scene	4281	-	99.79±0.09	99.80±0.06
	1800/3930	99.40±0.02	-	99.53±0.05

according to the literature, and the classification results are reported in Table XV. The results suggest that ASNN can achieve a better classification accuracy than 3-D-LWNet and SSRN with the same training samples.

In summary, we conducted three types of experiments in this section. First, we tested the effectiveness of spectral indices for RWCR, and RES exhibited the best performance. The proposed RES was applied to the inversion model for multilabel data generation by spectral index and adaptive grading algorithm. Second, we carried out a set of ablation studies of ASNN. The 11 × 11 spatial size was set as the default spatial size and the augmented multilabel data sets were set as default input of ASNN. Subsequently, we analyzed the classification improvements of two critical parts of ASNN. The results demonstrated that the SBS based on attention mechanism and data augmentation based on RWCR were both able to outperform the models without them, and ASNN combining with the two parts obtained the best classification accuracy. Third, we analyzed the refined classification performance of ASNN. The confusion matrix of classification results demonstrated that ASNN can improve the accuracy of intraclass and interclass in refined classification. Moreover, the classification accuracy of ASNN was able to be improved by

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the increase of the training samples, and ASNN was able to reach the best values of OA, AA, and Kappa coefficient with the lower standard deviation when compared with the state-of-the-art classifiers. It also suggested that ASNN achieved a quantitative and qualitative improvement with good stability in refined classification.

IV. Conclusion

In this article, a multisupervised neural network ASNN is proposed for HSI refined classification based on RWCR data augmentation. Specifically, the proposed ASNN is a deep learning model constrained by prior RWC information and requiring augmented multilabel data for input. The two symbiotic classification processes, RWC and spectral–spatial classification, are successively carried out in ASNN for biochemical–spectral–spatial feature extraction. There are three critical steps that need to be performed for the refined classification of ASNN. First, the original HSI data are demanded to be augmented and labeled with RWC information in the inversion model, which consists of the proposed spectral index RES calculation and adaptive grading algorithm. Second, the augmented multilabel data containing RWC and spectral–spatial information are taken as the input of ASNN. The attention-based SBS module in ASNN assigns the attention weights to all bands for the symbiotic classification processes. Third, the DVFE module in ASNN extracts RWC and spectral–spatial features successively and fuses them by parameter updating. It also enhances the synergetic spectral–spatial feature extraction by various feature maps. After training, ASNN is able to extract prior RWC and spectral–spatial features from multilabel data to perform refined classification. The experimental results demonstrate that the proposed RES is effective for RWCR, and the data augmentation based on RWCR inversion and SBS module based on attention mechanism both can improve the classification performance significantly. In addition, the superiority of ASNN for refined classification is also proved in the performance analysis and the comparison with state-of-the-art methods.

Although ASNN exhibits significant improvement, there are still some challenges that should be focused on. Particularly, the inherent properties of the exploration of data should be improved to develop more spectral indices so that multiple physical–chemical characteristics will be applied in refined classification. Moreover, the data-level fusion mode of RWC and spectral–spatial features can be improved to accelerate the training process and reduce the computational cost by various fusion pattern and simplified network.

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